

Name _____

Quiz 1

1. (2 points) For each symbol below, write its name or meaning in English. The first is filled in for you as an example.

- (a) $\mathcal{P}$: power set
(b) $\mathbb{R}$: _____
(c) $\cap$: _____
(d) $\cup$: _____
(e) $|$: _____

2. (2 points) Circle True or False for each of the following statements.

- (a) $a \in \{\{a\}, b\}$: True or False
(b) $\{1, 2\} \subseteq \{1, 2, 3, 5\}$: True or False
(c) $11 \in \{x \in \mathbb{Z} : x < 10\}$: True or False

3. (3 points) Given the following preference lists as input for the hospital/medical student matching problem.

H_1	:	S_3	S_1	S_2		S_1	:	H_1	H_2	H_3
H_2	:	S_3	S_2	S_1		S_2	:	H_3	H_2	H_1
H_3	:	S_2	S_3	S_1		S_3	:	H_1	H_2	H_3

- (a) Is the matching $M = \{(H_1, S_3), (H_2, S_1), (H_3, S_2)\}$ stable? Yes or No
(b) How do you know?

4. (3 points) The next question asks about the same preference lists, which are copied for your convenience.

H_1	:	S_3	S_1	S_2		S_1	:	H_1	H_2	H_3
H_2	:	S_3	S_2	S_1		S_2	:	H_3	H_2	H_1
H_3	:	S_2	S_3	S_1		S_3	:	H_1	H_2	H_3

- (a) Is the matching $M = \{(H_1, S_1), (H_2, S_3), (H_3, S_2)\}$ stable? Yes or No
(b) How do you know?